

Yoshitaka Inoue

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Machine Learning for Biological Perturbations and Therapeutic Response

University of Minnesota | Computer Science and Engineering

National Institutes of Health | NLM / NCI

inoue0426.github.io

Google Scholar | ORCID | GitHub

Research focus Machine learning for biological perturbations and therapeutic response; single-cell and functional genomics; intervention-aware representation learning; graph-based biomedical reasoning.

Research and Professional Experience

2024–present	National Library of Medicine / National Cancer Institute, NIH <i>Pre-doctoral Fellow</i>
	Developing machine learning methods for perturbation modeling and therapeutic response prediction across single-cell, pharmacogenomic, high-throughput screening, cancer cell-line, and patient datasets, including treatment-conditioned latent-state transition models and graph-based biomedical reasoning systems.
2022–2024	Google Summer of Code <i>Contributor (2022); Mentor (2023–2024)</i>
	Developed a drug-response prediction pipeline and mentored projects on PyTorch Geometric datasets and multi-tool LLM aggregation for biomedical analysis.
2019–2021	FreakOut, Inc. <i>Machine Learning Engineer</i>
	Applied machine learning to click-through-rate prediction in Tokyo, Japan.

Education

2021–present	University of Minnesota, Twin Cities <i>PhD Candidate, Computer Science and Engineering</i>
	Advisor: Prof. Rui Kuang; Co-advisor: Dr. Augustin Luna (NIH). Research on treatment-conditioned perturbation modeling, therapeutic response prediction, single-cell and functional genomics. Expected graduation: September 2027.
2017–2019	Nara Institute of Science and Technology <i>M.Eng., Information Science</i>
	MS/MS-based small-molecule prediction and annotation in collaboration with the UC Davis Genome Center and DataLab. Advisor: Prof. Shigehiko Kanaya.
Summer 2018	University of California, Davis <i>Visiting Scholar, Genome Center and DataLab</i>
	Collaborative research on MS/MS-based compound prediction and metabolomics data analysis. Advisors: Prof. Oliver Fiehn and Tobias Kind.
2012–2017	Kumamoto University <i>B.Sc., Biological Science</i>
	Experimental analysis of seagrass metabolism using molecular biology methods. Advisor: Prof. Susumu Takio.

Selected Publications

- Inoue, Y., Jeong, M., Hero, A., Kuang, R., & Luna, A.**
PerturbRx: Learning Treatment-Conditioned Latent Transitions for Patient Drug Response Prediction.
arXiv, 2026; submitted to ICLR 2027
- Inoue, Y., Song, T., Wang, X., Luna, A., & Fu, T.**
DrugAgent: Reliable Multi-Agent Aggregation under Conflicting Biomedical Evidence.
Scientific Reports, 2026, under revision
- Inoue, Y., Lee, H., Fu, T., Kuang, R., & Luna, A.**
drGT: Attention-Guided Gene Assessment of Drug Response Utilizing a Drug-Cell-Gene Heterogeneous Network.
BMC Bioinformatics, 2026
- Inoue, Y., Hao, N., Lu, Y., Fu, T., & Luna, A.**
Drug Discovery in the Era of Artificial Intelligence: From Target Identification to Clinical Trials.
Health Data Science, 2026, accepted
- Hoshi-Kadonosawa, Y., Reinhold, W. C., Taniyama, D., **Inoue, Y.**, et al.
TRPM4 Expression as a Predictive Biomarker and a Mechanistic Driver of Acetalax Activity in Prostate Cancer: Preclinical Efficacy Studies.
Molecular Cancer Therapeutics, 2026

Selected Publications (continued)

6. Elloumi, F., Reinhold, W. C., Varma, S., Wang, Y., Kinali, M., Arakawa, Y., Inoue, Y., et al.
CellMiner cross-database (CellMinerCDB) version 2.2 for explorations of patient-derived cancer cell line pharmacogenomics.
Nucleic Acids Research, 2026

7. Taniyama, D., Inoue, Y., Elloumi, F., Kim, Y. S., et al.
Spontaneous dynamic gene expression of Schlafen 11 (SLFN11) demonstrated by single-cell RNA sequencing in clonal cancer cell lines.
Oncogene, 2026, submitted

Conference and Workshop Papers

1. Inoue, Y., Song, T., Wang, X., Luna, A., & Fu, T.
DrugAgent: Reliable Multi-Agent Aggregation under Conflicting Biomedical Evidence.
ICML Workshop on Foundation Models for the Life Sciences (FM4LS), 2026

2. Inoue, Y., Fu, T., & Luna, A.
GraphPINE: Graph Importance Propagation for Interpretable Drug Response Prediction.
ICLR Workshop on Machine Learning for Genomics Explorations (MLGenX), 2025

3. Inoue, Y., Song, T., Wang, X., Luna, A., & Fu, T.
DrugAgent: Explainable Drug Repurposing Agent with Large Language Model-based Reasoning.
ICLR Workshop on Machine Learning for Genomics Explorations (MLGenX), 2025

Preprints and Research Software

1. Inoue, Y., Kulman, E., & Kuang, R.
BiGCN: Leveraging Cell and Gene Similarities for Single-cell Transcriptome Imputation with Bi-Graph Convolutional Networks.
bioRxiv, 2024

2. **OncoGraph** — open oncology knowledge graph and retrieval benchmark integrating drugs, genes, diseases, clinical trials, Gene Ontology terms, and provenance-linked biomedical evidence. github.com/inoue0426/OncoGraph

Selected Presentations and Honors

International Conference on Intelligent Systems for Molecular Biology (ISMB)	2026
Two accepted oral presentations: drGT in NetBio and DrugAgent in Text Mining; both also presented at the Student Council Symposium.	Washington, DC
26th Annual CCR-FYI Colloquium	2026
drGT oral presentation; selected as one of four Outstanding Postgraduate Fellow Finalists.	Rockville, MD
NIH Artificial Intelligence Symposium	2025
drGT oral presentation; selected among 5 of 78 listed abstracts.	Rockville, MD
• NIH Pre-Doctoral Research Fellowship , 2024–present.	
• Research Visit Scholarships , NAIST and JASSO support for UC Davis research visits, 2018–2019.	

Academic Service

Journal reviewing	BMJ Digital Health & AI (Expert Peer Reviewer, 2026–2027); IEEE/ACM Transactions on Computational Biology and Bioinformatics (2025–2026); Nature Computational Science (co-reviewer, 2026). ACM BCB (2023); ICML FM4LS (2026); ICML GenBio (2026); IEEE BIBM (2026); ACM SIGSPATIAL (2026).
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Conference reviewing

Technical Skills

ML / AI	Perturbation and transition modeling; graph neural networks; representation learning; LLM-based agents; interpretable machine learning.
Computational biology	Pharmacogenomics; functional genomics; biological networks; single-cell RNA-seq; therapeutic response modeling; multi-omics analysis.
Programming	Python; PyTorch; PyTorch Geometric; scikit-learn; pandas; NumPy.
Infrastructure	Git; Linux; Jupyter; Slurm; GPU computing; HPC environments; LangChain.